

BARC0130: Advanced Mathematical Modelling & Analysis

Coursework 2

Programming, Data Visualisation & Applied Linear Algebra Topics

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QUESTION 1

The NO₂ data collected near Euston Road was approximated with the Fourier series [1], both in MATLAB Task 1 and in equation 1 below, where the background and roadside NO₂ data were modelled as one and plotted in figures 2 to 5 [2][3]. This was done by using the AQM Excel data column labelled NO₂, which was assumed to be an accumulation of the roadside and background NO₂ emissions. The Fourier Series mathematical model was used to analyse the pattern of weekly NO₂ concentrations, such that the data was easy to understand for students and employees working in that area, especially since Euston Road is so close to the NO₂ hotspot of Europe (Marylebone Road) [4].

To summarise, the approach to fitting the data was to first clean the dataset, making sure to change the times in the original AQM Excel sheet to a manipulable array that could be plotted. Then the mean weekly pattern was calculated and plotted beside the overall data for comparison (see figures 2 and 3 below). Next was to fit the data to a Fourier series equation to model the typical weekly NO₂ patterns more cleanly (equation 1 and figure 4), and finally, to compute the relevant statistics and visualise them (equation 2, equation 3, table 1, table 2 and figure 5). The figures were spread throughout the paper so they would be a helpful visual aid to the NO₂ fitting methods. See figure 1 for the flow diagram.

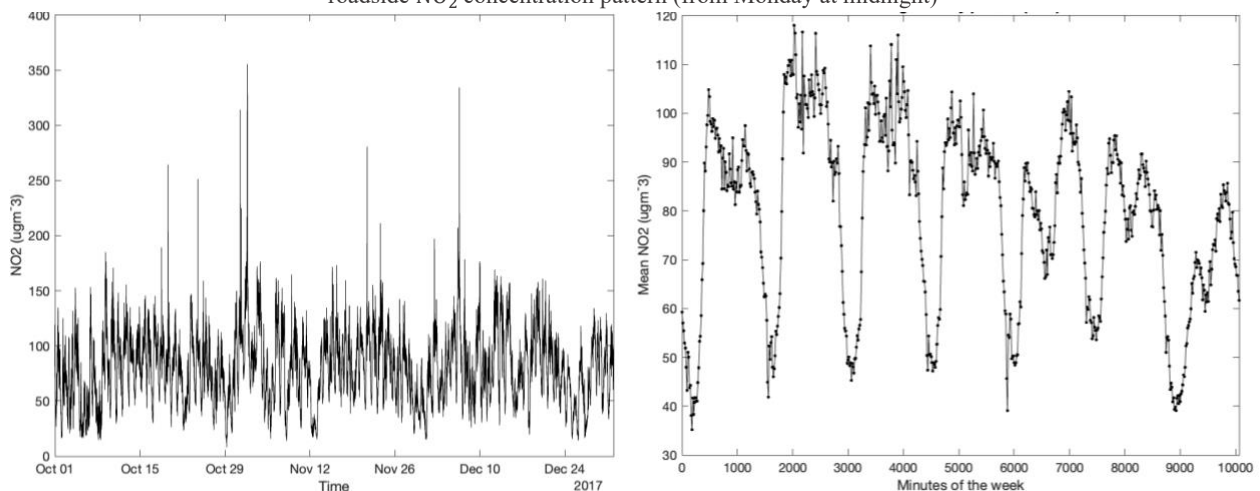
Figure 1. Flow diagram of the NO₂ data analysis process



The data in figure 2 shows an average NO₂ concentration of around 79.94 $\mu\text{g m}^{-3}$, which is well over the maximum annual average NO₂ concentration of 40 $\mu\text{g m}^{-3}$ set by the EU directive [5][6]. This is reflected in figure 3 where susceptible employees working in Bloomsbury could be facing respiratory problems when exposed to these levels of NO₂ [7].

The assumptions of the fitting method were that there was no meaningful pattern to the missing values in the dataset, and the quality of the fit wouldn't be reduced by removing them. The data from 1st October 2017 to 31st December 2017 was also assumed to be representative of the whole year. The limitations were that the data only captures an averaged week of NO₂ emissions and that different patterns that may appear over a greater time scale, were not considered (figure 3).

Figure 2. Cleaned background and roadside NO₂ concentration time series, Figure 3. Average weekly background and roadside NO₂ concentration pattern (from Monday at midnight)

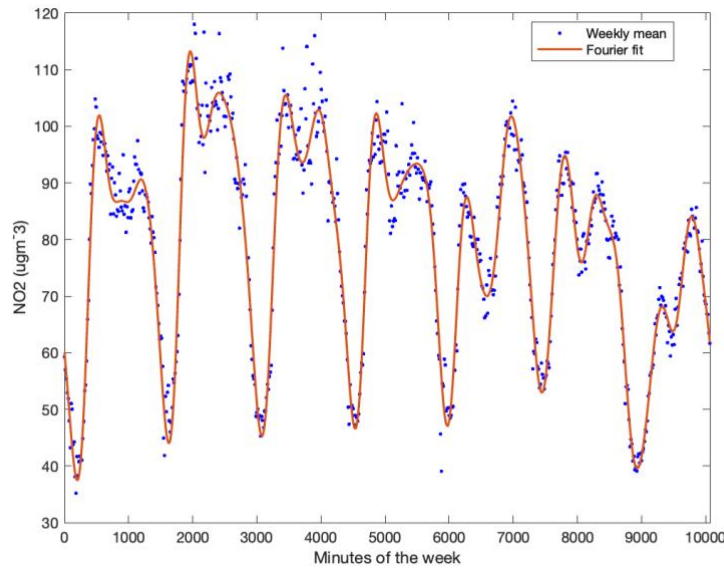


Equation 1 and figure 4 below show a 30-term Fourier approximation of the weekly NO₂ data (from figure 3). Unfortunately, this data shows that levels of NO₂ around Bloomsbury are not ideal for a workspace where people have to spend a lot of time. With an average background NO₂ concentration

of about $45.08 \mu\text{gm}^{-3}$ (exceeding the EU NO_2 limit) [6], the inhabitants may experience increased airflow obstruction or irritation in the lungs as well as reduced immunity to lung infections like bronchitis and pneumonia [7]. Additionally, since people spend about 30% of their weekdays in offices, reduced lung function due to long term exposure is a concern [7].

In order to circumvent these issues (using the fitted data in figure 4), it would be best to avoid working or staying in Bloomsbury, especially nearer the roads, from 8:30am to 7:30pm on weekdays where NO_2 levels are at a maximum ($90\text{--}110 \mu\text{gm}^{-3}$). Outside these times NO_2 is around $70 \mu\text{gm}^{-3}$ and can reach as low as $38 \mu\text{gm}^{-3}$ early Monday and Sunday morning. Overall, the weekend experiences average NO_2 concentrations around $60 \mu\text{gm}^{-3}$, which make these ideal working times in Bloomsbury.

Figure 4. Comparison of the mean weekly overall NO_2 pattern and the 30-term Fourier approximation (starts Monday at midnight)



In the MATLAB code for question 1, the Fourier series is calculated with the weights in mind (see MATLAB code for Q1) [3]. These weights (W) were imputed into the equation below and used to confirm the quality of the Fourier model later on (in table 1, table 2, equation 2, equation 3 and figure 5). The weights in equation 1 were calculated using MATLAB, where W_{ii} is equal to $\frac{1}{\sigma_i^2}$ is equal to $\frac{n_i}{s_i^2}$ when the weekly NO_2 data is a mean [1].

The Fourier Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega x + b_n \sin n\omega x) \quad n \in \mathbb{N} \text{ if harmonious}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \left(\frac{2\pi n}{10080} x \right) + b_n \sin \left(\frac{2\pi n}{10080} x \right) \right), \omega = \frac{2\pi}{T}, T = 10080$$

$$f(x) = f(x + T), \text{ where } T = 10080 \text{ mins (a week)}$$

Weighted Least Squares Fourier Analysis

$$W = \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_i^2} \end{bmatrix}, \quad W_{ii} = \frac{1}{\sigma_i^2} = \frac{n_i}{s_i^2} \quad (1)$$

$$W = \begin{bmatrix} 2.31e-2 & 0 & \cdots & 0 \\ 0 & 1.70e-2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 2.22e-2 \end{bmatrix} 2 \text{ d.p.}$$

$$Ax = b + v, W$$

$$\hat{x} = x = (A^T W A)^{-1} A^T W b$$

$$x = \left[\frac{a_o}{2} \quad a_1 \quad \cdots \quad a_{30} \quad b_1 \quad \cdots \quad b_{30} \right]^T, \quad a_o = 158.05 \mu g m^{-3} \text{ 2 d.p.}$$

Final Fourier Series Values

$$f(x) \approx \frac{a_o}{2} - \left(a_1 \cos\left(\frac{2\pi}{10080}x\right) + a_2 \cos\left(\frac{4\pi}{10080}x\right) + a_3 \cos\left(\frac{6\pi}{10080}x\right) - \right.$$

$$a_4 \cos\left(\frac{8\pi}{10080}x\right) - a_5 \cos\left(\frac{10\pi}{10080}x\right) \dots \left. + \left(b_1 \sin\left(\frac{2\pi}{10080}x\right) + b_2 \sin\left(\frac{4\pi}{10080}x\right) + \right.$$

$$b_3 \sin\left(\frac{6\pi}{10080}x\right) + b_4 \sin\left(\frac{8\pi}{10080}x\right) + b_5 \sin\left(\frac{10\pi}{10080}x\right) \dots \left. \right)$$

$$f(x) \approx 79.03 - \left(-5.85 \cos\left(\frac{2\pi}{10080}x\right) + 4.56 \cos\left(\frac{4\pi}{10080}x\right) + 1.68 \cos\left(\frac{6\pi}{10080}x\right) - \right.$$

$$2.55 \cos\left(\frac{8\pi}{10080}x\right) - 1.81 \cos\left(\frac{10\pi}{10080}x\right) \dots \left. + \left(6.25 \sin\left(\frac{2\pi}{10080}x\right) + \right.$$

$$2.56 \sin\left(\frac{4\pi}{10080}x\right) + 3.33 \sin\left(\frac{6\pi}{10080}x\right) + 1.31 \sin\left(\frac{8\pi}{10080}x\right) + \left.$$

$$0.63 \sin\left(\frac{10\pi}{10080}x\right) \dots \left. \right) \mu g m^{-3} \text{ 2 d.p.} \quad \text{Evaluated for 5 values of the 30 term series}$$

Where $f(x)$, a_o , a_n , b_n , n , T , ω , W_{ii} , W , n_i , s_i^2 , A , x , b and v or $\bar{v}$ the Fourier series, the Fourier coefficient a_o ($\mu g m^{-3}$), the Fourier coefficient a_n ($\mu g m^{-3}$), the Fourier coefficient b_n ($\mu g m^{-3}$), the n^{th} value in the series, the time period (minutes), the angular speed (rads/s), the weights at (i,i) ($\mu^{-2} g^{-2} m^6$), the weights of the mean data (each bin) ($\mu^{-2} g^{-2} m^6$), the i^{th} number of values, the i^{th} variance of the raw data ($\mu^2 g^2 m^{-6}$), the Fourier design matrix, the vector of estimated parameters ($\mu g m^{-3}$), the vector of mean NO₂ concentrations ($\mu g m^{-3}$) and the residuals ($\mu g m^{-3}$), respectively.

To assess the quality of my model, the weighted unit variance, formal 1σ uncertainties and residual statistics (table 1 and table 2) were calculated. See equations 2 and 3, as well as the Task 1 MATLAB code, for more information on how this was done.

Weighted Unit Variance Calculation

$$Ax = b + v, W \quad \text{or} \quad A\bar{x} = \bar{b} + \bar{v}, W$$

$$\therefore v = Ax - b$$

$$m > n \therefore \text{overdetermined sytem with more equations than unknowns} \quad (2)$$

$$\sigma_0^2 = \frac{v^T W v}{m - n}$$

$$\sigma_0^2 = 0.38 \mu^2 g^2 m^{-6} \text{ 2 d.p.}$$

Weighted Formal 1σ Uncertainties Calculation

$$C_x = \sigma_0^2 (A^T W A)^{-1} = \begin{bmatrix} 0.03 & -0.01 & \cdots & -1.88e-4 \\ -0.01 & 0.06 & \cdots & -3.13e-4 \\ \vdots & \vdots & \ddots & \vdots \\ -1.88e-4 & -3.13e-4 & \cdots & 0.05 \end{bmatrix} \text{ 2 d.p.} \quad (3)$$

$$\sigma_{x,i} = \sqrt{(C_x)_{ii}} = \begin{bmatrix} \sigma_{a_o/2} \\ \sigma_{a_1} \\ \vdots \\ \sigma_{b_{30}} \end{bmatrix} = \begin{bmatrix} 0.17 \\ 0.24 \\ \vdots \\ 0.22 \end{bmatrix} \mu g m^{-3} \text{ 2 d.p.}$$

Where, σ_0^2 , m , n , C_x , $\sigma_{x,i}$, $\sigma_{a_o/2}$, σ_{a_1} , σ_{b_5} , $(C_x)_{ii}$, $\bar{v}$, σ_v , a_i and b_i are the weighted unit variance ($\mu^2 g^2 m^{-6}$), the number of observations, the number of parameters, the covariance matrix ($\mu^2 g^2 m^{-6}$), the Formal 1σ uncertainty of parameter i ($\mu g m^{-3}$), the Formal 1σ uncertainty of $\frac{a_o}{2}$ ($\mu g m^{-3}$), the Formal 1σ uncertainty of a_1 ($\mu g m^{-3}$), the Formal 1σ uncertainty of b_5 ($\mu g m^{-3}$), the diagonal covariance values

at position $(i, i)(\mu^2 g^2 m^{-6})$, the mean of the residuals ($\mu g m^{-3}$), the standard deviation of the residuals ($\mu g m^{-3}$), the i^{th} a_n coefficient ($\mu g m^{-3}$) and the i^{th} b_n coefficient ($\mu g m^{-3}$), respectively.

In table 1 below, the residual mean ($-0.49 \mu g m^{-3}$) is relatively close to 0, indicating a good quality fit. The unit variance ($0.38 \mu^2 g^2 m^{-6}$) is also < 1 , meaning the weighted residuals are small in relation to the expected uncertainties. However, the standard deviation of the residuals ($\pm 4.01 \mu g m^{-3}$) and the formal 1σ uncertainties of the Fourier fit ($\approx 0.24 \mu g m^{-3}$) were slightly large in comparison to the NO_2 data, which could be a sign of slight inaccuracy within the Fourier fit. Overall, though, the 30-term Fourier fit seems to represent the data well and was therefore deemed a high quality estimation of the typical weekly pattern of NO_2 emissions around Euston Road.

Figure 5. Residuals of the 30-term Fourier fit for overall NO_2 concentrations (starts Monday at midnight)

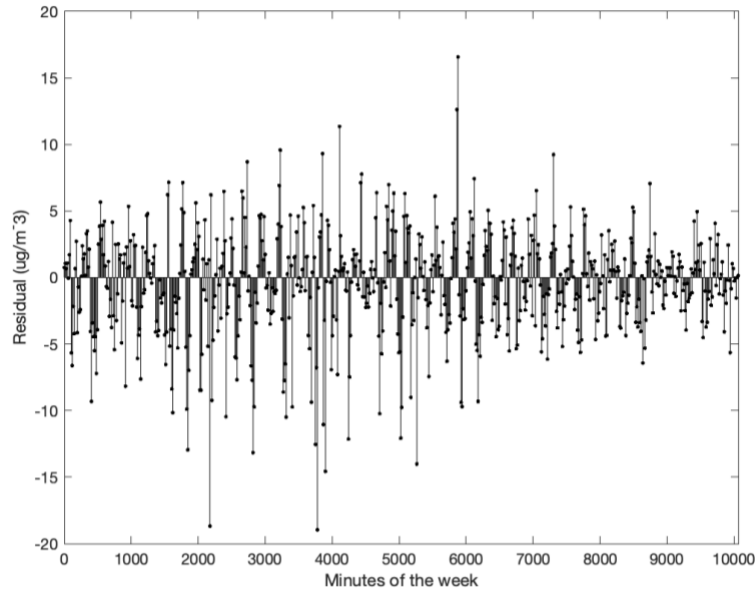


Table 1. Statistics value summary

Weighted Statistics	Values 2 d.p.
Unit variance, $\sigma_0^2 (\mu^2 g^2 m^{-6})$	0.38
Residual mean, $\bar{v} (\mu g m^{-3})$	-0.49
Residual standard deviation, $\sigma_v (\mu g m^{-3})$	± 4.01

Table 2. Statistics on the weighted 30-term Fourier series residuals for the first 5 values of a_n and b_n

Fourier coefficient	Value 2 d.p.	Formal Fourier 1σ uncertainties ($\mu g m^{-3}$) 2 d.p.
a_0	158.05	± 0.33
a_1	-5.85	± 0.24
a_2	-4.56	± 0.22
a_3	-1.68	± 0.24
a_4	2.55	± 0.24
a_5	1.81	± 0.23
$\vdots$	$\vdots$	$\vdots$
b_1	6.25	± 0.23
b_2	2.56	± 0.25
b_3	3.33	± 0.23
b_4	1.31	± 0.23
b_5	0.63	± 0.24

QUESTION 2A

In the equation below, the Excel building data was modelled into a series of matrices, which were used to calculate the least squares estimate of x , returning the unweighted unit cost of each material (see Task 2A MATLAB code for how this was done). Additionally, since the equation of the building materials was overdetermined, the least squares analysis method was used to get closest to satisfying the 25 equations by minimising the sum of the squared residuals (SSR)[8].

It's also worth noting that in the least squares analysis derivation, $(x^T A^T b)^T$ is equal to $b^T Ax$ because the terms are scalar (a 1x1 quantity) [9][10]. This enables the line $v^T v = x^T A^T Ax - b^T Ax - b^T Ax + b^T b$ to become $v^T v = x^T A^T Ax - 2b^T Ax + b^T b$ so that the equation can be further simplified and differentiated [4][11].

Unweighted Unit Cost Calculation

$G\bar{c}_g + C\bar{c}_c + S\bar{c}_s = \text{Cost}$, for the first row of the Q2 building data

$$\therefore Ax = b$$

$$\therefore \begin{bmatrix} G_1 & C_1 & S_1 \\ G_2 & C_2 & S_2 \\ G_3 & C_3 & S_3 \\ \vdots & \vdots & \vdots \\ G_{25} & C_{25} & S_{25} \end{bmatrix} \begin{bmatrix} \bar{c}_g \\ \bar{c}_c \\ \bar{c}_s \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_{25} \end{bmatrix} \therefore \begin{bmatrix} 22864 & 142642 & 104890 \\ 18502 & 118137 & 81186 \\ 11022 & 71376 & 51301 \\ \vdots & \vdots & \vdots \\ 11562 & 79682 & 56129 \end{bmatrix} \begin{bmatrix} \bar{c}_g \\ \bar{c}_c \\ \bar{c}_s \end{bmatrix} = \begin{bmatrix} 69905520 \\ 54716876 \\ 34307482 \\ \vdots \\ 37671151 \end{bmatrix}$$

Least Squares Derivation

$$Ax = b + v \quad \text{or} \quad \underline{Ax} = \bar{b} + \bar{v}$$

$$\therefore v = Ax - b, \text{ which is the linear model of the data}$$

Minimising the SSR (Unweighted)

$$\min_x \|Ax - b\|^2$$

$$\|Ax - b\|^2 = (Ax - b)^T (Ax - b) \quad (4)$$

$$\therefore v^T v = (Ax - b)^T (Ax - b)$$

$$v^T v = x^T A^T Ax - b^T Ax - b^T Ax + b^T b$$

$$\therefore v^T v = x^T A^T Ax - 2b^T Ax + b^T b,$$

$$\text{where } (x^T A^T b)^T = b^T Ax \text{ (terms are scalar)}$$

Differentiate

$$\frac{\partial v^T v}{\partial x} = 2A^T Ax - 2A^T b = 0, \quad \text{equate with 0 to solve for a minimum}$$

$$\therefore A^T Ax = A^T b, \quad \text{multiply both sides by } (A^T A)^{-1} \text{ to get:}$$

$$\hat{x} = x = (A^T A)^{-1} A^T b$$

Least Squares Estimate of x ($\hat{x}$)

$$x = \begin{bmatrix} \bar{c}_g \\ \bar{c}_c \\ \bar{c}_s \end{bmatrix} = £ \begin{bmatrix} 37.81 \\ 75.13 \\ 556.42 \end{bmatrix} 2 \text{ d. p.}$$

Where G , C , S , $\bar{c}_g$, $\bar{c}_c$, $\bar{c}_s$, A , x , b , c , v or $\bar{v}$ and $\hat{x}$ are the amount of glass in m^2 , the amount of concrete in m^3 , the amount of steel in tons, the unit cost of glass (£ m^{-2}), the unit cost of concrete (£ m^{-3}), the unit cost of steel (£ ton^{-1}), the material quantity matrix, the matrix of the unit costs, the matrix of the total costs, the scalar total cost values within matrix b (£), the residuals (£) and the least squares estimate of x , respectively.

In the equation below, the unit variance (5.10 e + 8, 2 d. p.) is a measure of how well the unweighted model fits the building cost data under the assumption that the buildings are equally reliable (see

equation 5 and the Task 2A MATLAB code for how this was calculated) [11]. Since the unit variance is so large, the unweighted model may be a poor fit for the data. In equation 6, the unweighted formal uncertainties can also be found as the square root of the diagonals in the covariance matrix.

Unweighted Unit Variance Calculation

$$\begin{aligned}
 Ax &= b + v \quad \text{or} \quad \underline{A}\bar{x} = \bar{b} + \bar{v} \\
 \therefore v &= Ax - b, \text{ which is the linear model of the data} \\
 m > n &\therefore \text{overdetermined system with more equations than unknowns} \\
 \sigma_0^2 &= \frac{v^T v}{m - n} \\
 \sigma_0^2 &= 5.10 e + 8 \quad 2 \text{ d.p.}
 \end{aligned} \tag{5}$$

Unweighted Formal 1σ Uncertainties Calculation

$$\begin{aligned}
 C_x &= \sigma_0^2 (A^T A)^{-1} = \begin{bmatrix} 62.88 & -4.92 & -6.72 \\ -4.92 & 1.25 & -0.70 \\ -6.72 & -0.70 & 2.47 \end{bmatrix} \quad 2 \text{ d.p.} \\
 \sigma_{1,i} &= \sqrt{(C_x)_{ii}} = \begin{bmatrix} 7.93 \\ 1.12 \\ 1.57 \end{bmatrix} \quad 2 \text{ d.p.}
 \end{aligned} \tag{6}$$

Where σ_0^2 , m , n , C_x , $\sigma_{1,i}$, and $(C_x)_{ii}$ are the unweighted unit variance (£^2), the number of observations, the number of parameters, the covariance matrix, the formal 1σ uncertainties at column i and the diagonal covariance values at position (i, i) , respectively.

Using the values calculated in equations 4 to 6, the unweighted unit cost of glass, concrete and steel are as below (see in table 3). The unweighted 1σ uncertainty for glass is the largest (at 21% of the unit cost), while the 1σ uncertainties for concrete and steel are relatively well constrained at 1.5% and 0.3%, respectively. The mean of the residuals ($-\text{£}43.72 \quad 2 \text{ d.p.}$) is approximately 0 (relative to data), which, along with the relatively small standard deviation of the residuals ($\pm \text{£}2.16 e + 4 \quad 2 \text{ d.p.}$), suggests that despite flaws with the unit costs of glass, the unweighted fit has some merit (see table 4 and figure 6).

Figure 6. A plot of the unweighted residuals

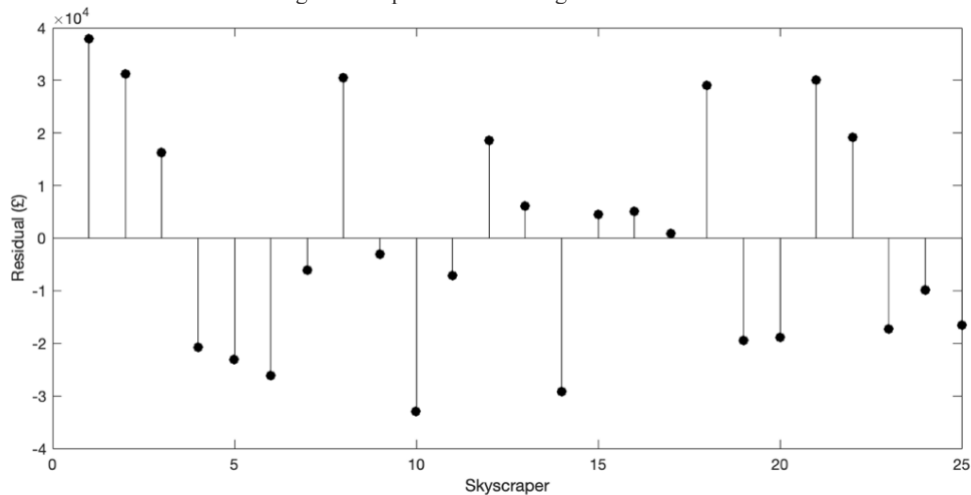


Table 3. The unweighted unit costs and formal 1σ uncertainties of glass, concrete and steel

Material	Unweighted unit costs (£) 2 d.p.	Unweighted formal 1σ uncertainty 2 d.p.
Glass (m^2)	37.81	± 7.93
Concrete (m^3)	75.13	± 1.12

Steel (tons ²)	556.42	±1.57
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Table 4. The mean and standard deviation of the unweighted residuals

Data	Mean, $\bar{v}_1$ (£) 2 d.p.	Standard deviation, σ_1 (£) 2 d.p.
Unweighted Residuals	- 43.72	± 2.16 e + 4

QUESTION 2B

Below (eq. 7-9) are the weighted calculations of the unit cost, unit variance and formal 1σ uncertainties (see Task 2B MATLAB code and Q2A above for more detail on how these values were calculated).

Weighted Unit Cost Calculation

$$W = \begin{bmatrix} 7.95 e - 10 & 0 & \dots & 0 \\ 0 & 8.54 e - 10 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1.87 e - 9 \end{bmatrix} 2 d.p.$$

Least Squares Derivation

$$\begin{aligned} v^T W v &= (Ax - b)^T W (Ax - b) \\ \therefore v^T W v &= x^T A^T W A x - 2b^T W A x + b^T W b \\ \frac{\partial v^T W v}{\partial x} &= 2A^T W A x - 2A^T W b = 0, \quad \text{equate with 0 to solve for a minimum} \\ \therefore A^T W A x &= A^T W b, \quad \text{multiply both sides by } (A^T W A)^{-1} \text{ to get:} \\ \hat{x} = x &= (A^T W A)^{-1} A^T W b \end{aligned} \quad (7)$$

Least Squares Estimate of x ($\hat{x}$)

$$x = \begin{bmatrix} \bar{c}_g \\ \bar{c}_c \\ \bar{c}_s \end{bmatrix} = \begin{bmatrix} 43.97 \\ 74.35 \\ 556.18 \end{bmatrix} 2 d.p.$$

Weighted Unit Variance Calculation

$$\begin{aligned} Ax &= b + v, \quad W \quad \text{or} \quad A\bar{x} = \bar{b} + \bar{v}, \quad W \\ \therefore v &= Ax - b, \quad \text{which is the linear model of the data} \\ m > n &\therefore \text{overdetermined system with more equations than unknowns} \\ \sigma_0^2 &= \frac{v^T W v}{m - n} \\ \sigma_0^2 &= £^2 0.93 \quad 2 d.p. \end{aligned} \quad (8)$$

Weighted Formal 1σ Uncertainties Calculation

$$\begin{aligned} C_x &= \sigma_0^2 (A^T W A)^{-1} = \begin{bmatrix} 1.67 & -0.30 & 0.06 \\ -0.30 & 0.22 & -0.26 \\ 0.06 & -0.26 & 0.36 \end{bmatrix} 2 d.p. \\ \sigma_{1,i} &= \sqrt{(C_x)_{ii}} = \begin{bmatrix} 1.29 \\ 0.47 \\ 0.60 \end{bmatrix} 2 d.p. \end{aligned} \quad (9)$$

Where W and σ_0^2 are the weight of the variables (£⁻²) and the weighted unit variance (£²), respectively.

Using the values from equations 7 to 9, the calculation for the weighted unit variance ($\text{£}^2 \ 0.93 \ 2 \text{ d. p.}$) reveals a significantly smaller number than the unweighted unit variance ($5.10 \ e + 8, 2 \text{ d. p.}$). This, along with the weighted formal 1σ uncertainties (see table 5) being much smaller than their unweighted counterparts (2.9%, 0.63% and 0.11% uncertainties for glass, concrete and steel, respectively), suggests that the weighted unit costs (which can be found in table 5) are more accurate than the unweighted values. Figure 7 and table 6 also show that the mean ($\text{£}205.77 \ 2 \text{ d. p.}$) and standard deviation ($\pm \text{£}2.20 \ e + 4, 2 \text{ d. p.}$) of the weighted residuals are not that much different from unweighted values. However, despite this, it is clear that the weighted model is the more statistically appropriate fit for the data (especially for the unit cost of glass) [9][12].

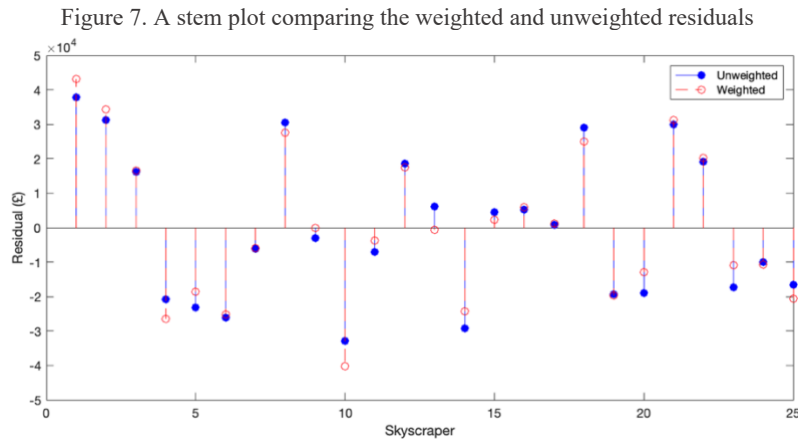


Table 5. The weighted unit costs and formal 1σ uncertainties of glass, concrete and steel

Material	Weighted unit costs (£) 2 d.p.	Weighted formal 1σ uncertainty 2 d.p.
Glass (m^2)	43.97	± 1.29
Concrete (m^3)	74.35	± 0.47
Steel ($tons^2$)	556.18	± 0.60

Table 6. The mean and standard deviation of the weighted residuals

Data	Mean, $\bar{v}_2$ (£) 2 d.p.	Standard deviation, σ_2 (£) 2 d.p.
Weighted Residuals	205.77	$\pm 2.20 \ e + 4$

QUESTION 2C

Below is the calculation of the estimated material costs for the 100-floor skyscraper where the weighted unit cost was used such that the final answer would be as accurate as possible. The code in task 2C was used to calculate the total cost of the skyscraper, where the final value was found as $\text{£}67,080,000 \ 4 \text{ s.f.}$

$$\begin{aligned}
 A_2 x &= b_2 \\
 \therefore b_2 &= A_2 x, \text{ where } x \text{ is the weighted unit cost} \\
 b_2 &= [21800 \quad 141230 \quad 100000] \begin{bmatrix} 43.97 \\ 74.35 \\ 556.18 \end{bmatrix} = \text{£}6.71 \ e + 7 \ 2 \text{ d. p.} \\
 b_2 &= 958,500 + 10,500,000 + 55,620,000 \ 4 \text{ s. f.} \\
 b_2 &= \text{£}67,080,000 \ 4 \text{ s. f.}
 \end{aligned} \tag{10}$$

Where A_2 and b_2 are the new material quantity matrix and the total cost of the new skyscraper, respectively.

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